

# Select and construct appropriate representations, including tables, graphs, models and mathematical relationships, to organise and process data and information

## Learning intention

- select and construct appropriate representations, including tables, graphs, models and mathematical relationships, to organise and process data and information.

## Teach and model

- using simple formulas in spreadsheets to organise and process collected data.
- constructing graphs using correct conventions such as naming the graph and labelling the axes.
- using visual displays of large data sets, such as maps showing the location of volcanoes and earthquakes.

## Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

## Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

## Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.